

Project name : “Final Project”

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Course: HDS 5310

**Project Purpose:** The purpose of this project is to examine how environmental and social determinants—including air pollution (e.g., PM2.5), land development, and socioeconomic conditions—are associated with lung cancer incidence across U.S. counties.

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```
#load the neccesary libraries
```

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2     3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## * dplyr::filter() masks stats::filter()
## * dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
```

```
##
## The following object is masked from 'package:dplyr':
##
##   recode
##
## The following object is masked from 'package:purrr':
##
##   some
```

```
library(lmtest)
```

```
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(readxl)
library(dplyr)
library(ggplot2)
```

```
#load the dataset
lung.cancer.incidence <- read_excel("C:/Users/drhud/Downloads/Data_Set_Final_LTD_Slope_
Intercept (1).xlsx")
```

```
#View the first few rows of the dataset to get an overview of the structure and values
head(lung.cancer.incidence)
```

```
## # A tibble: 6 × 34
##   FIPS_code County   State `Lung Cancer` PM2.5 `Status Variable` Land_EQI
##   <dbl> <chr>    <chr>      <dbl> <dbl>          <dbl>    <dbl>
## 1    1001 Autauga   AL          73.9  12.1            1    -0.707
## 2    1003 Baldwin  AL          68.4  11.1            1    -1.08
## 3    1005 Barbour  AL          76.1  12.4            1    -1.28
## 4    1007 Bibb Cou AL          86.4  12.2            1    -0.827
## 5    1009 Blount C AL          73.1  13.0            1    -0.623
## 6    1011 Bullock AL          72.4  12.2            1    -1.26
## # i 27 more variables: Sociod_EQI <dbl>, Built_EQI <dbl>, LTD <dbl>,
## #   Intercept <dbl>, Slope <dbl>, CLU50_1 <dbl>, PM10 <dbl>, SO2 <dbl>,
## #   NO2 <dbl>, O3 <dbl>, CO <dbl>, CN <dbl>, Diesel <dbl>, CS2 <dbl>,
## #   Air_EQI <dbl>, Water_EQI <dbl>, EQI <dbl>, Status <dbl>, LCI <dbl>,
## #   UCI <dbl>, Inter <dbl>, Slp <dbl>, control <dbl>, treat <dbl>,
## #   Local_Treat <dbl>, AAC <dbl>, RT <chr>
```

```
#View the last few rows of the dataset to check for any trailing NA values or inconsistencies
tail(lung.cancer.incidence)
```

```
## # A tibble: 6 × 34
##   FIPS_code County      State `Lung Cancer` PM2.5 `Status Variable` Land_EQI
##   <dbl> <chr>      <chr>      <dbl> <dbl>      <dbl>      <dbl>
## 1    56033 Sheridan WY          49.9  7.57          0    0.283
## 2    56037 Sweetwat WY          45.3  6.4           0    0.329
## 3    56039 Teton Co WY          22.6  6.67          0    0.317
## 4    56041 Uinta Co WY          37.1  6.57          0    0.253
## 5    56043 Washakie WY         51.1  6.33          0    0.151
## 6    56045 Weston C WY         46.4  5.24          0    0.298
## # i 27 more variables: Sociod_EQI <dbl>, Built_EQI <dbl>, LTD <dbl>,
## #   Intercept <dbl>, Slope <dbl>, CLU50_1 <dbl>, PM10 <dbl>, SO2 <dbl>,
## #   NO2 <dbl>, O3 <dbl>, CO <dbl>, CN <dbl>, Diesel <dbl>, CS2 <dbl>,
## #   Air_EQI <dbl>, Water_EQI <dbl>, EQI <dbl>, Status <dbl>, LCI <dbl>,
## #   UCI <dbl>, Inter <dbl>, Slp <dbl>, control <dbl>, treat <dbl>,
## #   Local_Treat <dbl>, AAC <dbl>, RT <chr>
```

```
#Generate summary statistics for each variable
summary(lung.cancer.incidence)
```

```
##   FIPS_code      County      State      Lung Cancer
##   Min.      :1001    Length:2602    Length:2602    Min.      : 12.90
##   1st Qu.:18028    Class :character    Class :character    1st Qu.: 58.00
##   Median :31002    Mode  :character    Mode  :character    Median : 68.60
##   Mean      :30741                                Mean      : 69.17
##   3rd Qu.:46081                                3rd Qu.: 79.40
##   Max.      :56045                                Max.      :169.90
##
##   PM2.5      Status Variable      Land_EQI      Sociod_EQI
##   Min.      : 1.700    Min.      :0.0000    Min.      : -5.11581    Min.      : -4.809990
##   1st Qu.: 8.453    1st Qu.:0.0000    1st Qu.: -0.40068    1st Qu.: -0.649570
##   Median :10.650    Median :1.0000    Median : 0.17488    Median : 0.004630
##   Mean      :10.125    Mean      :0.5058    Mean      : 0.03327    Mean      : 0.009265
##   3rd Qu.:11.860    3rd Qu.:1.0000    3rd Qu.: 0.64774    3rd Qu.: 0.617387
##   Max.      :16.910    Max.      :1.0000    Max.      : 2.09453    Max.      : 3.979472
##
##   Built_EQI      LTD      Intercept      Slope
##   Min.      : -3.99272    Min.      : -22.44    Min.      : -32.97    Min.      : -1.150
##   1st Qu.: -0.43898    1st Qu.: 5.86    1st Qu.: 32.46    1st Qu.: 2.250
##   Median : 0.17925    Median : 10.52    Median : 35.42    Median : 3.150
##   Mean      : 0.07986    Mean      : 12.29    Mean      : 38.22    Mean      : 3.068
##   3rd Qu.: 0.68986    3rd Qu.: 15.26    3rd Qu.: 45.59    3rd Qu.: 3.440
##   Max.      : 3.88379    Max.      : 50.64    Max.      : 89.59    Max.      :10.540
##
##   CLU50_1      PM10      SO2      NO2
##   Min.      : 1.00    Min.      : 1.220    Min.      : 1.00    Min.      : 1.012
##   1st Qu.:13.00    1st Qu.: 8.155    1st Qu.: 44.65    1st Qu.: 234.953
##   Median :24.00    Median :11.530    Median : 143.10    Median : 457.056
##   Mean      :25.21    Mean      :11.905    Mean      : 233.54    Mean      : 592.969
##   3rd Qu.:40.00    3rd Qu.:14.770    3rd Qu.: 307.14    3rd Qu.: 779.318
##   Max.      :50.00    Max.      :39.550    Max.      :12180.53    Max.      :8661.627
##
##   O3      CO      CN      Diesel
```

```
## Min.      : 1.64    Min.      : 1.112    Min.      :0.0001281    Min.      :0.01688
## 1st Qu.: 2895.29    1st Qu.: 185.524    1st Qu.:0.0071636    1st Qu.:0.19244
## Median : 4574.93    Median : 356.301    Median :0.0154060    Median :0.29696
## Mean    : 5126.00    Mean    : 604.817    Mean    :0.0242717    Mean    :0.39313
## 3rd Qu.: 6594.43    3rd Qu.: 952.495    3rd Qu.:0.0304405    3rd Qu.:0.46681
## Max.     :80276.76    Max.     :24815.739    Max.     :1.3493422    Max.     :8.81512
##
##          CS2          Air_EQI          Water_EQI          EQI
## Min.      :0.0000000    Min.      :-2.8190    Min.      :-1.64127    Min.      :-3.2200
## 1st Qu.:0.0001230    1st Qu.: -0.2887    1st Qu.: -0.89906    1st Qu.: -0.4600
## Median :0.0003230    Median : 0.2303    Median : 0.25090    Median : 0.1300
## Mean     :0.0054496    Mean     : 0.2078    Mean     : 0.04251    Mean     : 0.1223
## 3rd Qu.:0.0009447    3rd Qu.: 0.7765    3rd Qu.: 0.90958    3rd Qu.: 0.7500
## Max.     :2.3000000    Max.     : 2.7898    Max.     : 1.47818    Max.     : 2.8500
##
##          Status          LCI          UCI          Inter
## Min.      :1.000    Min.      : 9.20    Min.      : 17.50    Min.      : -32.97
## 1st Qu.:1.000    1st Qu.: 46.23    1st Qu.: 69.90    1st Qu.: 30.25
## Median :2.000    Median : 58.30    Median : 82.50    Median : 36.81
## Mean     :1.506    Mean     : 57.19    Mean     : 83.97    Mean     : 42.53
## 3rd Qu.:2.000    3rd Qu.: 67.50    3rd Qu.: 96.20    3rd Qu.: 50.25
## Max.     :2.000    Max.     :138.60    Max.     :224.40    Max.     :255.86
##                                     NA's      :2552
##          Slp          control          treat          Local_Treat
## Min.      :-70.630    Min.      :39.70    Min.      :40.84    Min.      : -22.440
## 1st Qu.: 1.695    1st Qu.:58.80    1st Qu.:70.62    1st Qu.: 4.284
## Median : 2.905    Median :62.46    Median :76.39    Median : 8.531
## Mean     : 1.543    Mean     :64.12    Mean     :74.50    Mean     :10.849
## 3rd Qu.: 3.647    3rd Qu.:69.90    3rd Qu.:78.29    3rd Qu.:15.086
## Max.     :10.540    Max.     :99.80    Max.     :96.09    Max.     :50.638
## NA's      :2552    NA's      :2552    NA's      :2554    NA's      :2554
##          AAC          RT
## Min.      : 3.00    Length:2602
## 1st Qu.: 14.00    Class :character
## Median : 28.00    Mode  :character
## Mean     : 76.18
## 3rd Qu.: 66.00
## Max.     :3650.00
##
```

```
#Select relevant variables for analysis from the dataset
# These include lung cancer incidence, air pollution metrics (PM2.5, PM10, NO2, SO2, O3, CO),
# environmental quality indices (Air_EQI, Sociod_EQI, Built_EQI), and region type (RT)
lung.cancer <- lung.cancer.incidence %>%
  select("Lung Cancer", "PM2.5", "PM10", "NO2", "SO2", "O3", "CO", "Air_EQI", "Sociod_EQI", "Built_EQI", "RT")
```

```
# Check the structure of the selected data frame to confirm variable types and names
str(lung.cancer)
```

```
## tibble [2,602 × 11] (S3: tbl_df/tbl/data.frame)
## $ Lung_Cancer: num [1:2602] 73.9 68.4 76.1 86.4 73.1 72.4 58.5 79.6 72.5 85.9 ...
## $ PM2.5      : num [1:2602] 12.1 11.1 12.4 12.2 13 ...
## $ PM10       : num [1:2602] 15.1 20 15.8 14.9 17.9 ...
## $ NO2        : num [1:2602] 123.7 247.7 183.2 127.8 95.2 ...
## $ SO2        : num [1:2602] 10.66 17.15 23.26 7.63 8.91 ...
## $ O3         : num [1:2602] 522 541 896 563 562 ...
## $ CO         : num [1:2602] 4.46 12.88 19.62 2.95 9.36 ...
## $ Air_EQI    : num [1:2602] 0.9554 0.718 0.131 0.0653 0.4022 ...
## $ Sociod_EQI : num [1:2602] 0.67 0.553 -1.236 -0.6 0.297 ...
## $ Built_EQI  : num [1:2602] -0.4973 0.4016 0.0489 -1.2909 -1.2627 ...
## $ RT        : chr [1:2602] "stable" "stable" "stable" "stable" ...
```

```
# Convert Region Type (RT) to a factor, and ensure all other selected variables are numeric
# This step is important for consistent data types before statistical analysis
lung.cancer <- lung.cancer %>%
  mutate(
    RT = as.factor(RT),
    `Lung Cancer` = as.numeric(`Lung Cancer`),
    `PM2.5` = as.numeric(`PM2.5`),
    PM10 = as.numeric(PM10),
    NO2 = as.numeric(NO2),
    SO2 = as.numeric(SO2),
    O3 = as.numeric(O3),
    CO = as.numeric(CO),
    Air_EQI = as.numeric(Air_EQI),
    Sociod_EQI = as.numeric(Sociod_EQI),
    Built_EQI = as.numeric(Built_EQI)
  )
```

```
# Recheck the structure to confirm successful type conversions
str(lung.cancer)
```

```
## tibble [2,602 × 11] (S3: tbl_df/tbl/data.frame)
## $ Lung_Cancer: num [1:2602] 73.9 68.4 76.1 86.4 73.1 72.4 58.5 79.6 72.5 85.9 ...
## $ PM2.5      : num [1:2602] 12.1 11.1 12.4 12.2 13 ...
## $ PM10       : num [1:2602] 15.1 20 15.8 14.9 17.9 ...
## $ NO2        : num [1:2602] 123.7 247.7 183.2 127.8 95.2 ...
## $ SO2        : num [1:2602] 10.66 17.15 23.26 7.63 8.91 ...
## $ O3         : num [1:2602] 522 541 896 563 562 ...
## $ CO         : num [1:2602] 4.46 12.88 19.62 2.95 9.36 ...
## $ Air_EQI    : num [1:2602] 0.9554 0.718 0.131 0.0653 0.4022 ...
## $ Sociod_EQI : num [1:2602] 0.67 0.553 -1.236 -0.6 0.297 ...
## $ Built_EQI  : num [1:2602] -0.4973 0.4016 0.0489 -1.2909 -1.2627 ...
## $ RT        : Factor w/ 4 levels "*", "falling", ...: 4 4 4 4 4 4 4 4 2 4 ...
```

```
# Remove rows with invalid region type indicated by "*"
# This step ensures data quality by removing potential place holder or undefined categories
lung.cancer <- lung.cancer %>%
```

```
filter(RT != "*")
```

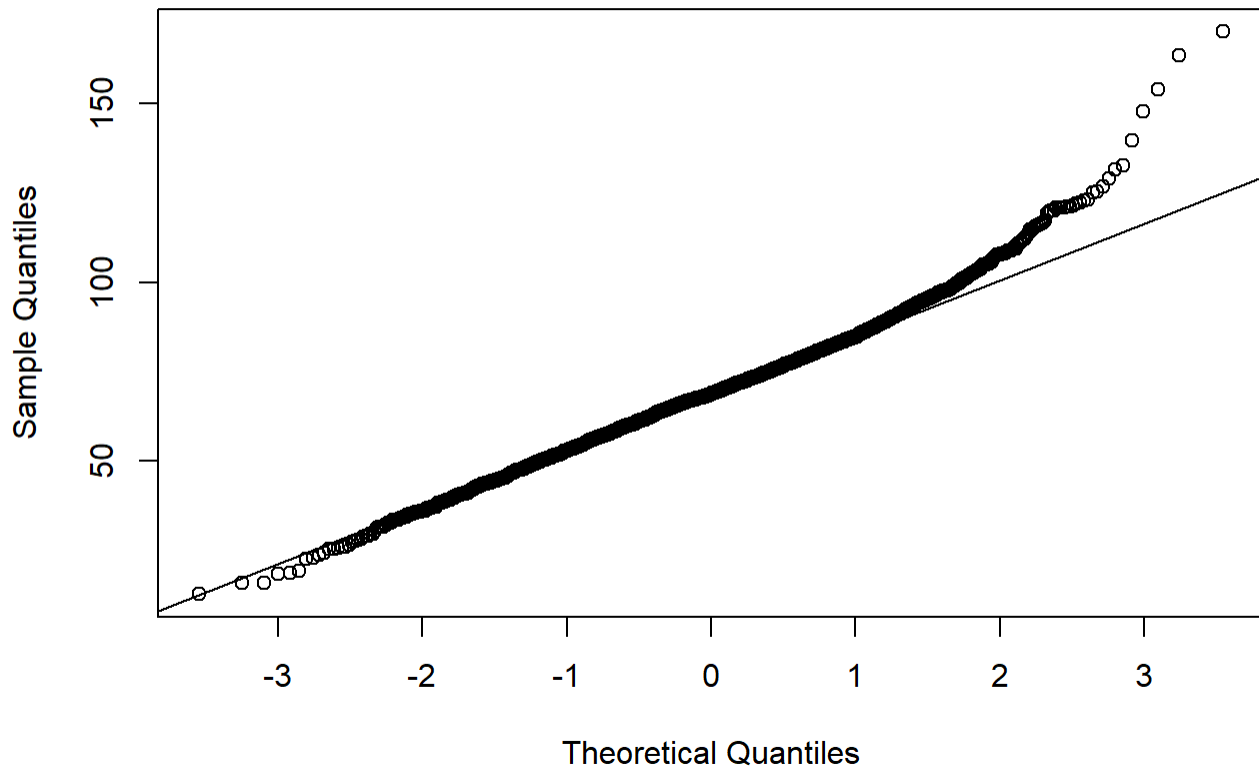
```
lung.cancer$RT <- droplevels(lung.cancer$RT)
```

```
# Summarize the cleaned dataset  
summary(lung.cancer)
```

```
##      Lung Cancer      PM2.5      PM10      NO2  
## Min.   : 12.90   Min.   : 1.70   Min.   : 1.22   Min.   : 1.012  
## 1st Qu.: 58.25   1st Qu.: 8.55   1st Qu.: 8.24   1st Qu.: 237.955  
## Median : 68.80   Median :10.68   Median :11.55   Median : 461.510  
## Mean   : 69.45   Mean   :10.17   Mean   :11.96   Mean   : 598.188  
## 3rd Qu.: 79.60   3rd Qu.:11.86   3rd Qu.:14.79   3rd Qu.: 785.390  
## Max.   :169.90   Max.   :16.91   Max.   :39.55   Max.   :8661.627  
##      SO2      O3      CO      Air_EQI  
## Min.   : 1.00   Min.   : 1.64   Min.   : 1.112   Min.   : -2.8190  
## 1st Qu.: 46.65   1st Qu.: 2930.07   1st Qu.: 188.106   1st Qu.: -0.2593  
## Median : 145.45   Median : 4608.37   Median : 359.111   Median : 0.2470  
## Mean   : 236.53   Mean   : 5161.04   Mean   : 610.486   Mean   : 0.2337  
## 3rd Qu.: 312.17   3rd Qu.: 6626.22   3rd Qu.: 966.005   3rd Qu.: 0.7881  
## Max.   :12180.53   Max.   :80276.76   Max.   :24815.739   Max.   : 2.7898  
##      Sociod_EQI      Built_EQI      RT  
## Min.   : -4.809990   Min.   : -3.99272   falling: 237  
## 1st Qu.: -0.647421   1st Qu.: -0.42593   rising : 51  
## Median : 0.007954   Median : 0.18809   stable :2267  
## Mean   : 0.015968   Mean   : 0.08832  
## 3rd Qu.: 0.626615   3rd Qu.: 0.69234  
## Max.   : 3.979472   Max.   : 3.88379
```

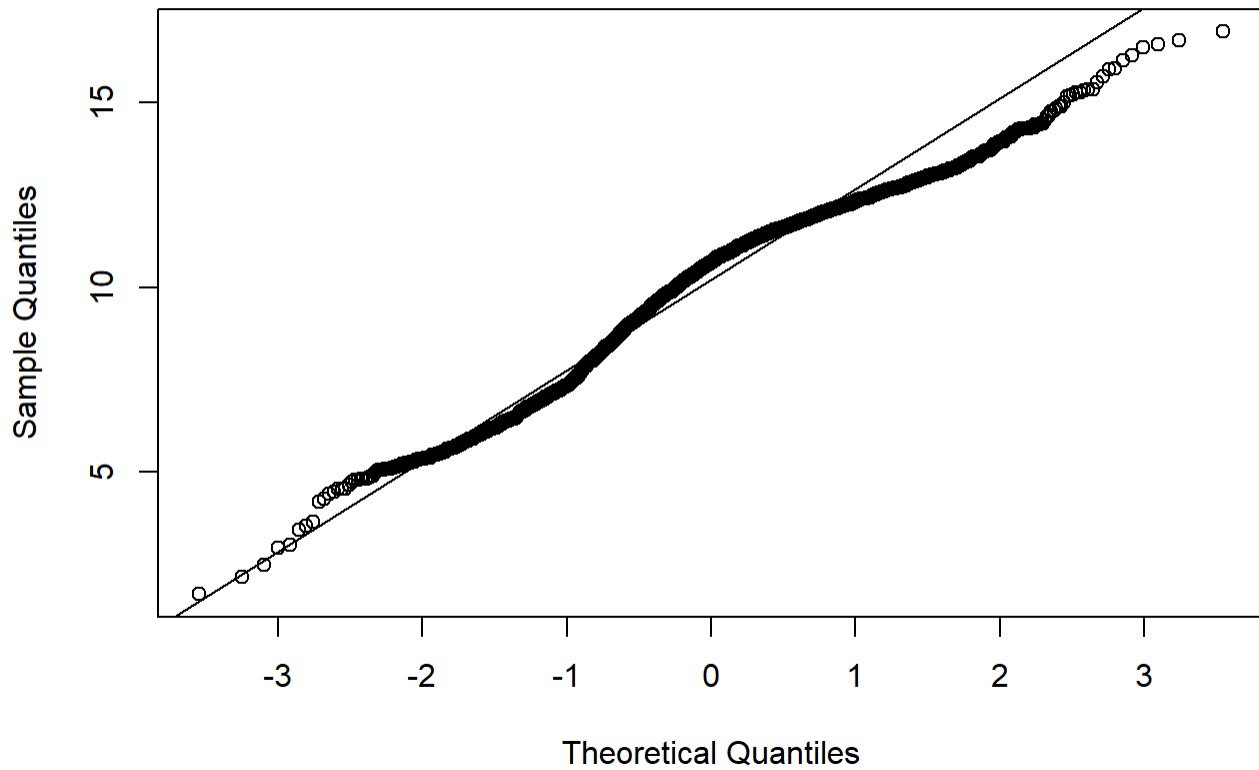
```
# Generate Q-Q plots to visually assess the normality of continuous variables.  
# If the data points roughly follow the reference line, the variable is approximately n  
ormally distributed.  
# Normality helps determine which descriptive statistics to use:  
# If normally distributed: use mean and standard deviation (SD)  
# If not normally distributed: use median and interquartile range (IQR)  
  
# Q-Q plot for Lung Cancer  
qqnorm(lung.cancer$`Lung Cancer`, main = "Q-Q Plot: Lung Cancer Incidence")  
qqline(lung.cancer$`Lung Cancer`)
```

## Q-Q Plot: Lung Cancer Incidence



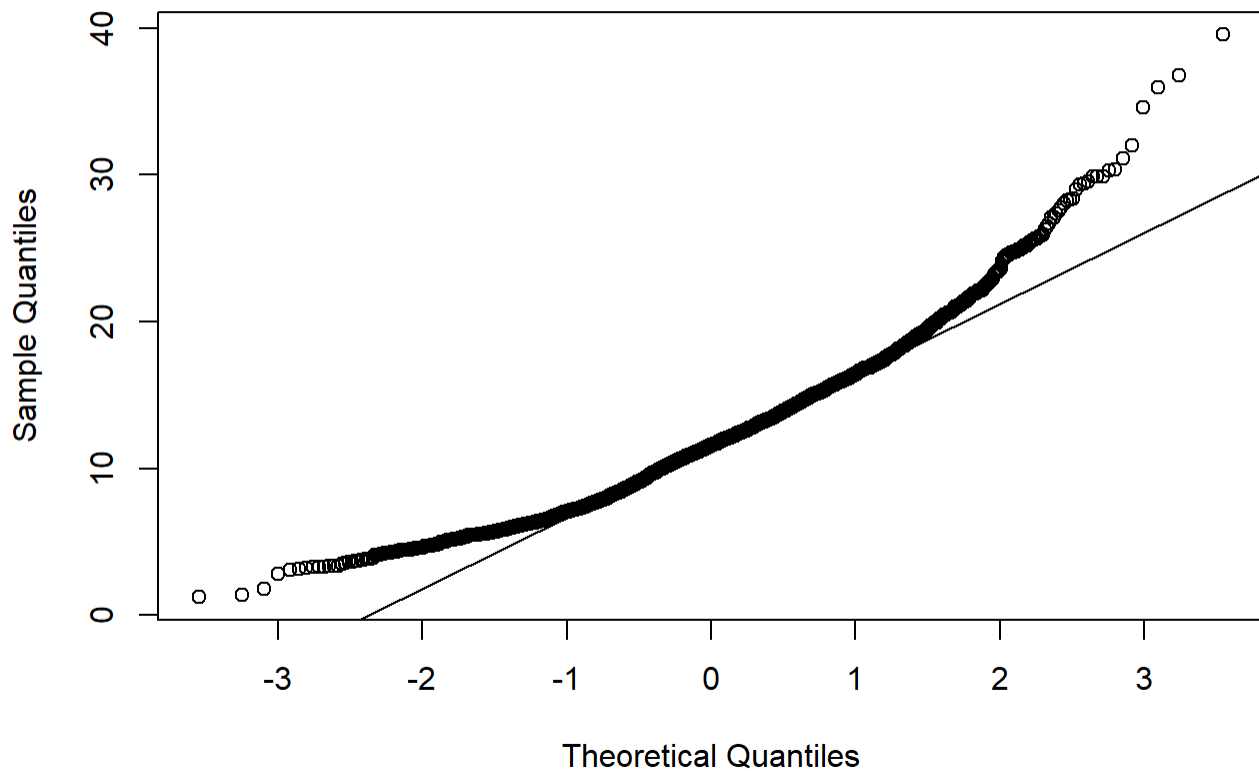
```
# Q-Q plot for PM2.5  
qqnorm(lung.cancer$`PM2.5`, main = "Q-Q Plot: PM2.5")  
qqline(lung.cancer$`PM2.5`)
```

## Q-Q Plot: PM2.5



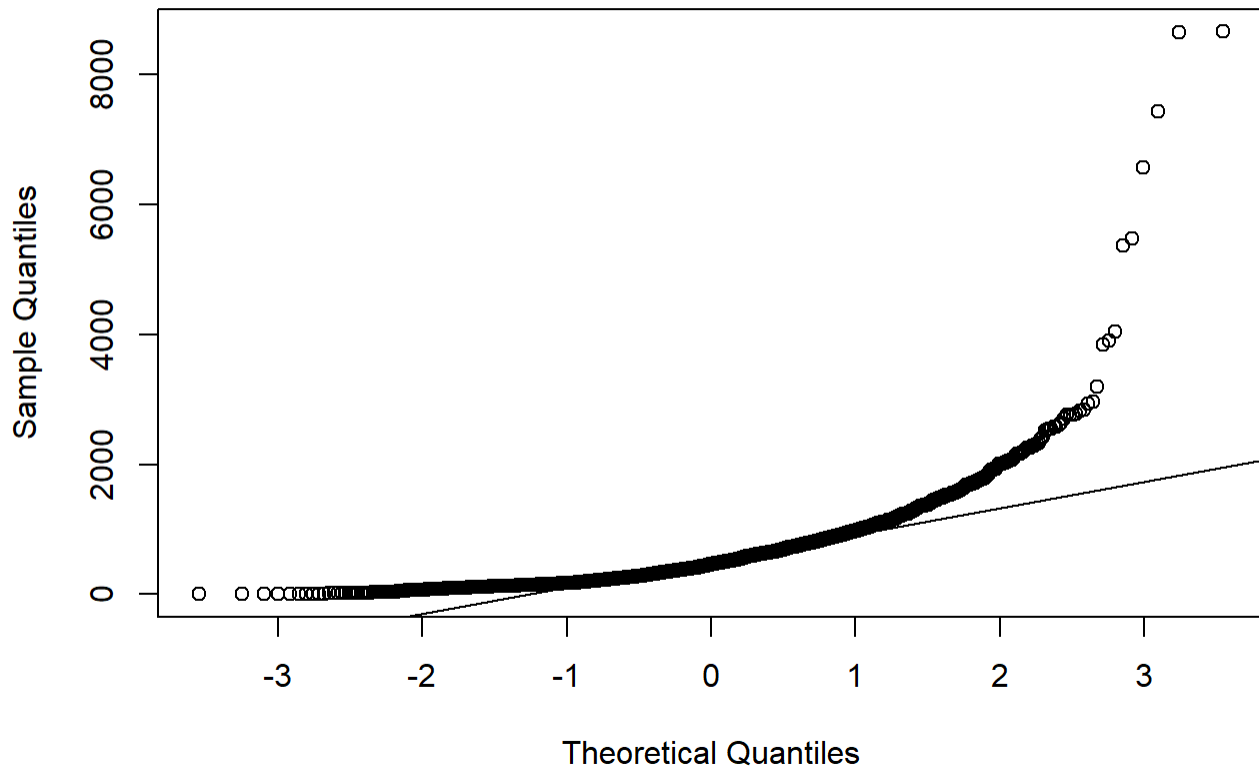
```
# Q-Q plot for PM10  
qqnorm(lung.cancer$PM10, main = "Q-Q Plot: PM10")  
qqline(lung.cancer$PM10)
```

## Q-Q Plot: PM10



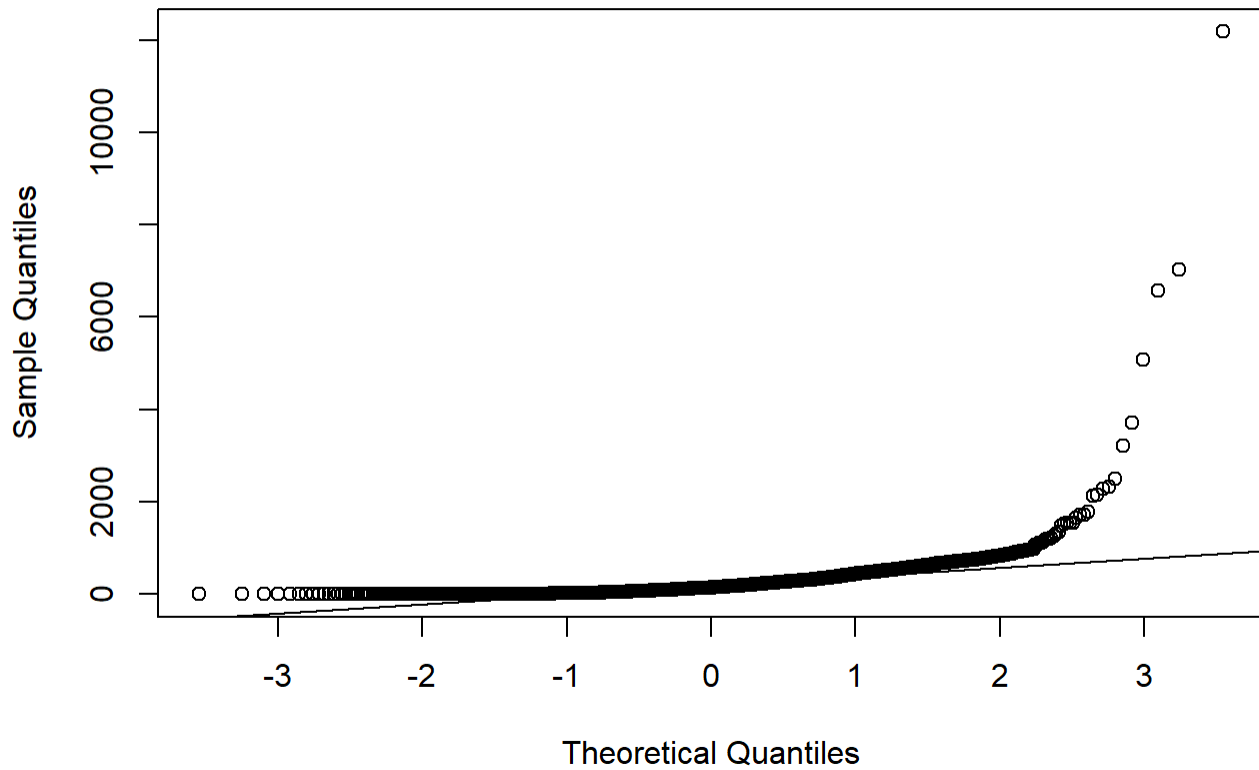
```
# Q-Q plot for N02  
qqnorm(lung.cancer$N02, main = "Q-Q Plot: N02")  
qqline(lung.cancer$N02)
```

## Q-Q Plot: NO2



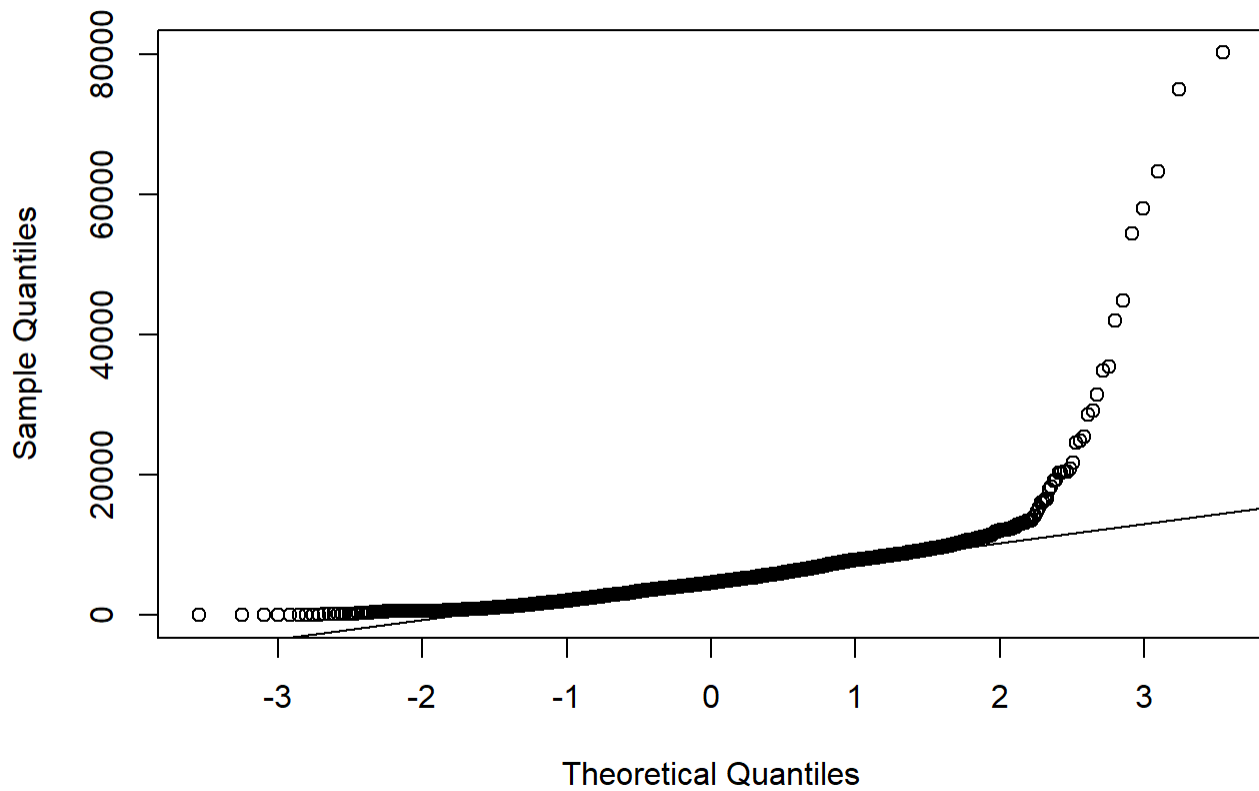
```
# Q-Q plot for S02  
qqnorm(lung.cancer$S02, main = "Q-Q Plot: S02")  
qqline(lung.cancer$S02)
```

## Q-Q Plot: SO2



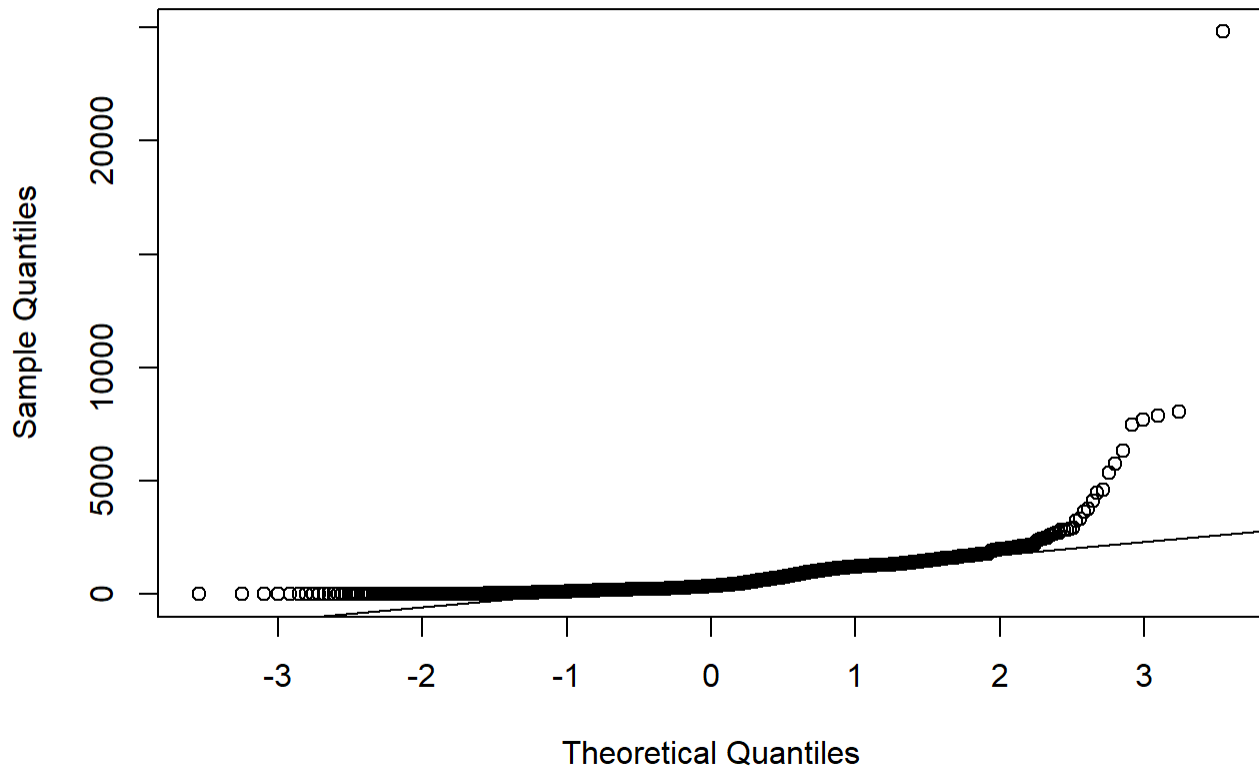
```
# Q-Q plot for 03  
qqnorm(lung.cancer$03, main = "Q-Q Plot: 03")  
qqline(lung.cancer$03)
```

### Q-Q Plot: O3



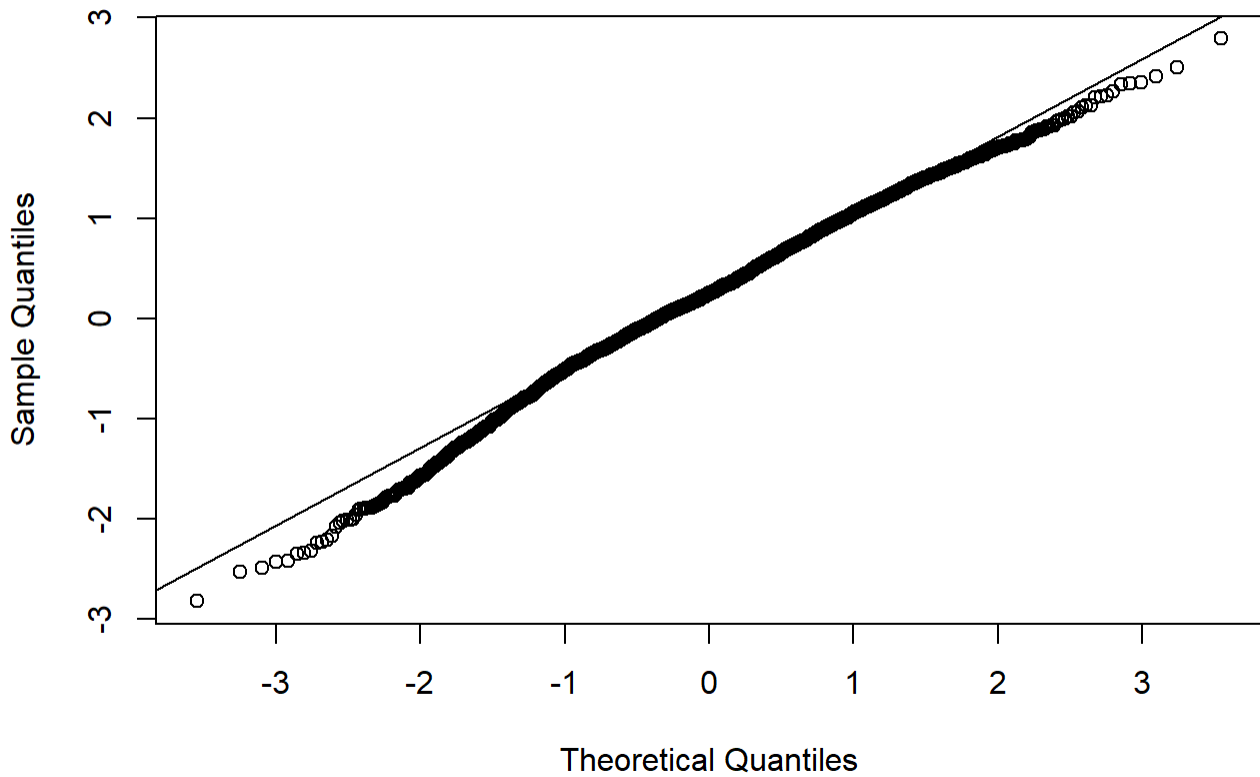
```
# Q-Q plot for CO  
qqnorm(lung.cancer$CO, main = "Q-Q Plot: CO")  
qqline(lung.cancer$CO)
```

## Q-Q Plot: CO



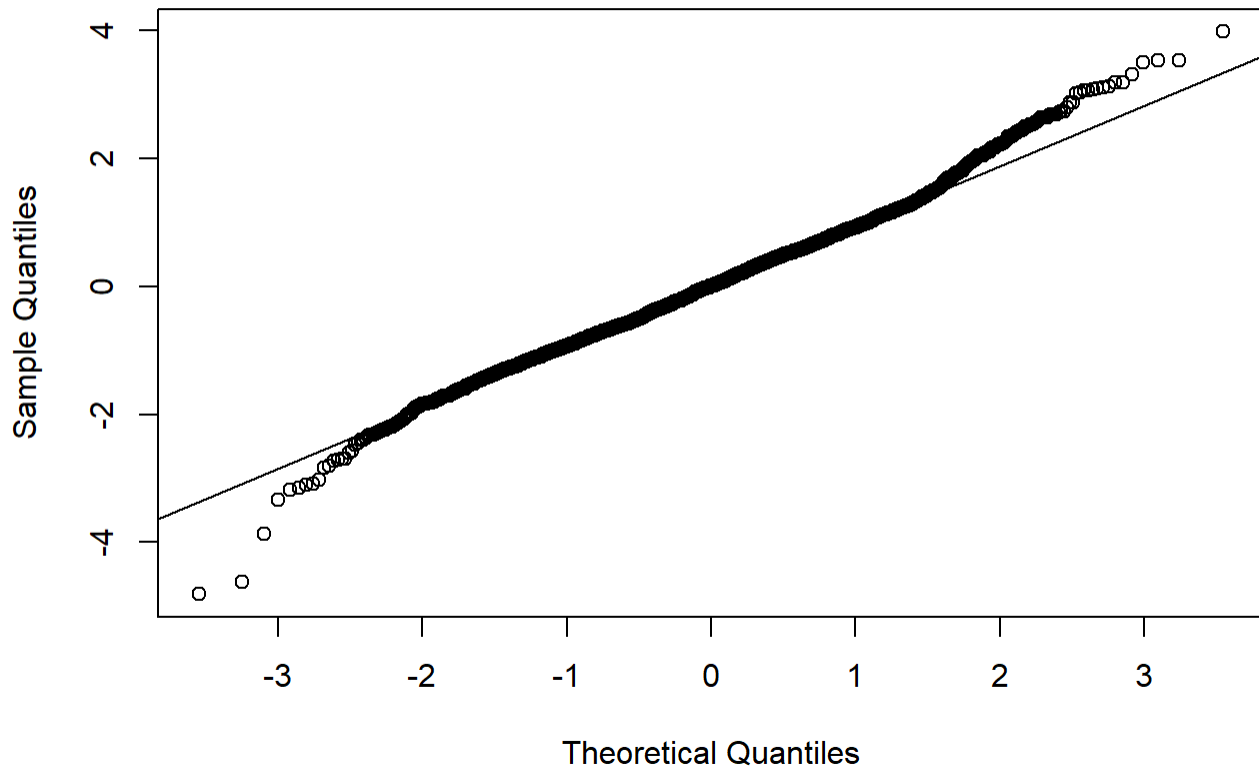
```
# Q-Q plot for Air_EQI  
qqnorm(lung.cancer$Air_EQI, main = "Q-Q Plot: Air EQI")  
qqline(lung.cancer$Air_EQI)
```

### Q-Q Plot: Air EQI



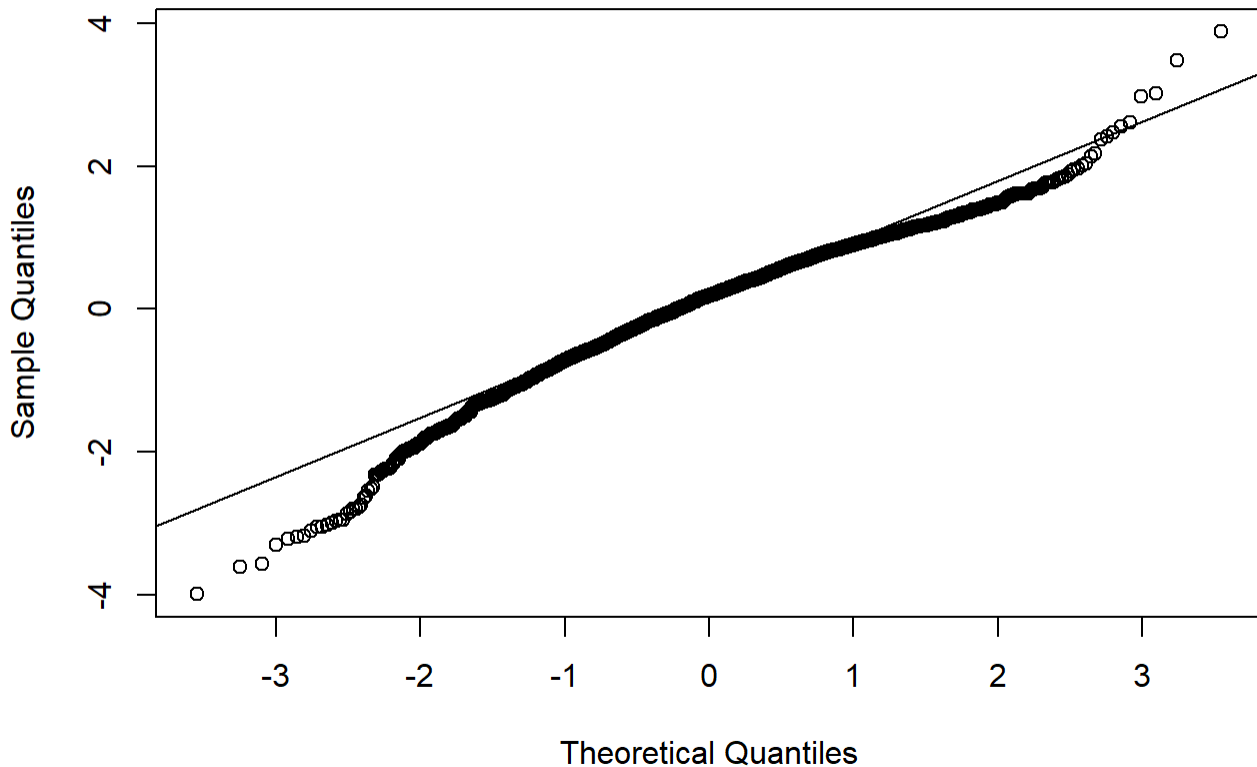
```
# Q-Q plot for Sociod_EQI  
qqnorm(lung.cancer$Sociod_EQI, main = "Q-Q Plot: Sociodemographic EQI")  
qqline(lung.cancer$Sociod_EQI)
```

## Q-Q Plot: Sociodemographic EQI



```
# Q-Q plot for Built_EQI
qqnorm(lung.cancer$Built_EQI, main = "Q-Q Plot: Built EQI")
qqline(lung.cancer$Built_EQI)
```

## Q-Q Plot: Built EQI



```
# Calculate summary statistics (mean & SD or median & IQR based on distribution)
lung_summary <- lung.cancer %>%
  summarise(
    mean_lung_cancer = mean(`Lung Cancer`, na.rm = TRUE),
    sd_lung_cancer = sd(`Lung Cancer`, na.rm = TRUE),

    mean_pm2_5 = mean(`PM2.5`, na.rm = TRUE),
    sd_pm2_5 = sd(`PM2.5`, na.rm = TRUE),

    mean_pm10 = mean(PM10, na.rm = TRUE),
    sd_pm10 = sd(PM10, na.rm = TRUE),

    mean_so2 = mean(SO2, na.rm = TRUE),
    sd_so2 = sd(SO2, na.rm = TRUE),

    mean_o3 = mean(O3, na.rm = TRUE),
    sd_o3 = sd(O3, na.rm = TRUE),

    mean_co = mean(CO, na.rm = TRUE),
    sd_co = sd(CO, na.rm = TRUE),

    median_no2 = median(NO2, na.rm = TRUE),
    iqr_no2 = IQR(NO2, na.rm = TRUE),

    mean_air_eq_i = mean(Air_EQI, na.rm = TRUE),
    sd_air_eq_i = sd(Air_EQI, na.rm = TRUE),
```

```

mean_sociod_eqi = mean(Sociod_EQI, na.rm = TRUE),
sd_sociod_eqi = sd(Sociod_EQI, na.rm = TRUE),

mean_built_eqi = mean(Built_EQI, na.rm = TRUE),
sd_built_eqi = sd(Built_EQI, na.rm = TRUE)
)

lung_summary

```

```

## # A tibble: 1 × 20
##   mean_lung_cancer sd_lung_cancer mean_pm2_5 sd_pm2_5 mean_pm10 sd_pm10 mean_so2
##   <dbl>          <dbl>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1      69.4        17.3        10.2     2.31     12.0     4.84    237.
## # i 13 more variables: sd_so2 <dbl>, mean_o3 <dbl>, sd_o3 <dbl>, mean_co <dbl>,
## #   sd_co <dbl>, median_no2 <dbl>, iqr_no2 <dbl>, mean_air_eqi <dbl>,
## #   sd_air_eqi <dbl>, mean_sociod_eqi <dbl>, sd_sociod_eqi <dbl>,
## #   mean_built_eqi <dbl>, sd_built_eqi <dbl>

```

```

# Frequency count of Region Type (RT)
table(lung.cancer$RT)

```

```

##
## falling rising stable
##      237      51    2267

```

```

# Percentage distribution of Region Type (RT)
prop.table(table(lung.cancer$RT)) * 100

```

```

##
## falling rising stable
## 9.275930 1.996086 88.727984

```

## BIVARIATE ANALYSIS: PM2.5 and Lung Cancer Incidence:

#Assumptions for Pearson Correlation:

ASSUMPTION 1 - Independence of Observations :Each row in the dataset represents a unique county. There are no repeated measures or clustering, so this ASSUMPTION is MET.

ASSUMPTION 2- Both Variables are Continuous : PM2.5 (particulate matter concentration) and Lung Cancer incidence are continuous variables. This ASSUMPTION is MET.

ASSUMPTION 3 - Normal Distribution of Both Variables: We assess this using Q-Q plots for both PM2.5 and Lung Cancer incidence.

```

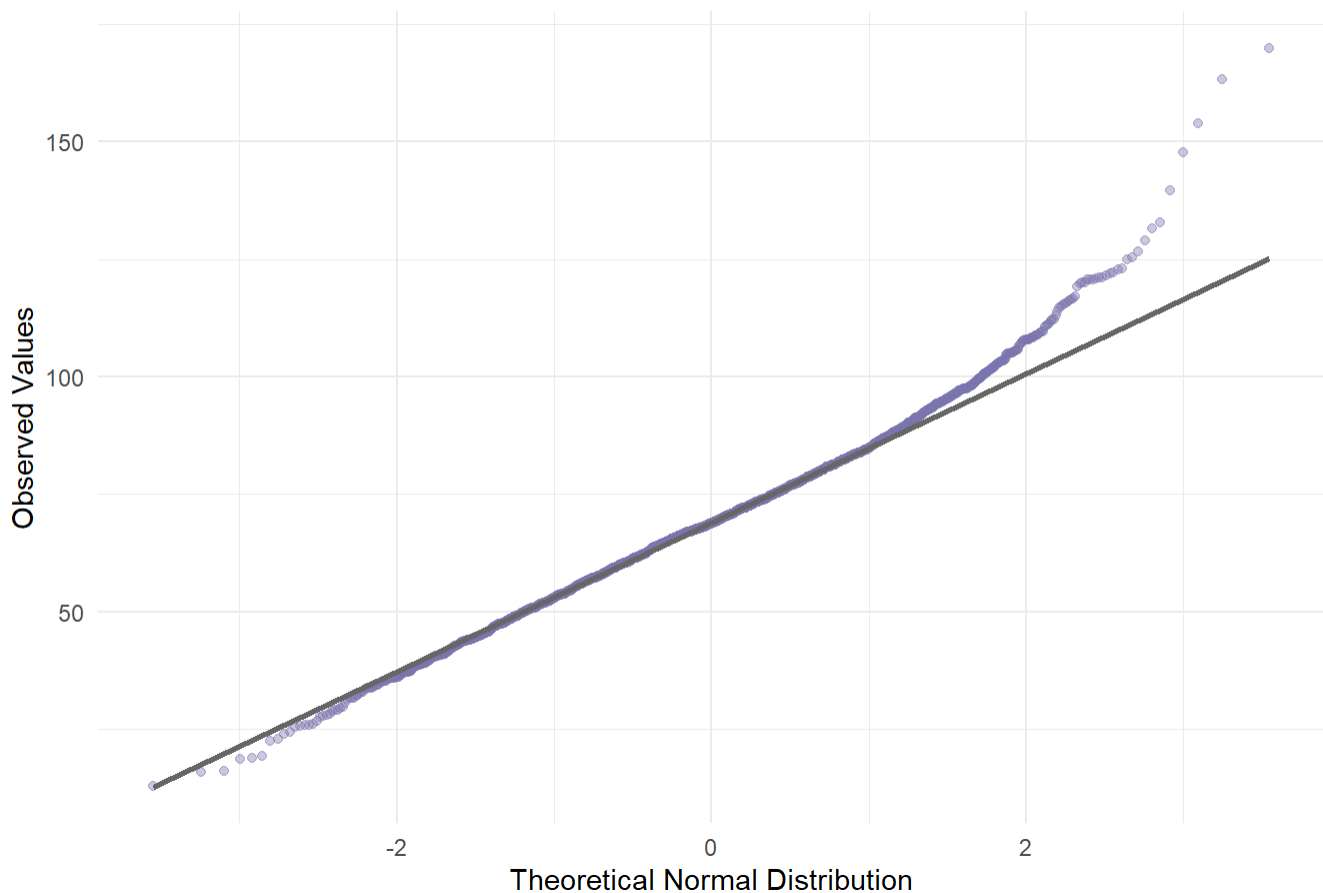
# Q-Q plot to assess normality of Lung Cancer Incidence
ggplot(lung.cancer, aes(sample = `Lung Cancer`)) +
  stat_qq(alpha = 0.4, color = "#7A74AC") +
  stat_qq_line(color = "gray40", size = 1) +

```

```
labs(
  title = "Q-Q Plot of Lung Cancer Incidence",
  x = "Theoretical Normal Distribution",
  y = "Observed Values"
) +
theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Q-Q Plot of Lung Cancer Incidence

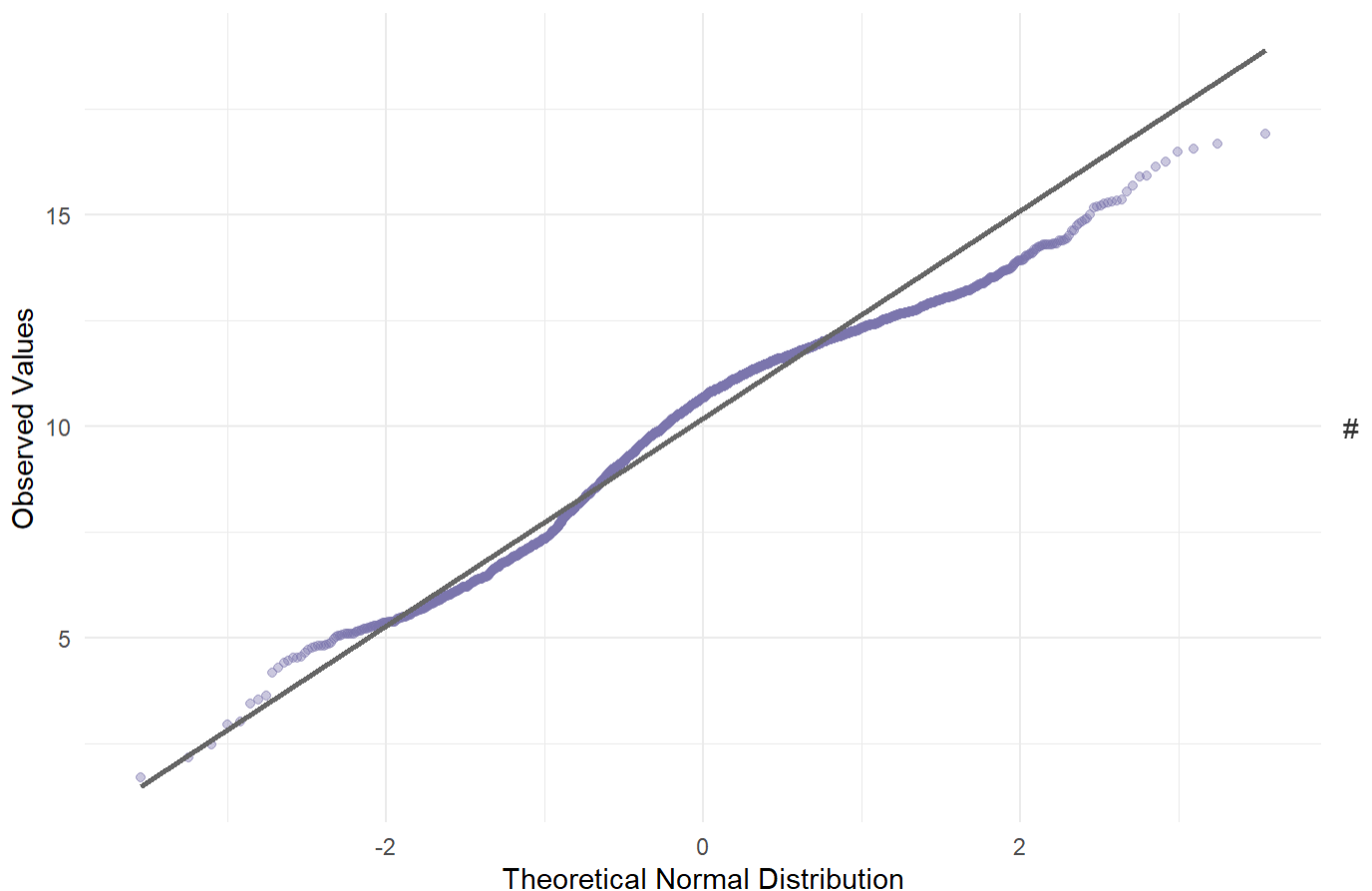


**#Interpretation of Q-Q Plot: Lung Cancer Incidence** The Q-Q plot shows that most of the data points for lung cancer incidence fall closely along the reference line, with only mild deviations at the upper end. This indicates that the variable is approximately normally distributed, thereby meeting the assumption of normality required for parametric tests such as the Pearson correlation.

```
# Q-Q plot to assess normality of PM2.5
ggplot(lung.cancer, aes(sample = `PM2.5`)) +
  stat_qq(alpha = 0.4, color = "#7A74AC") +
  stat_qq_line(color = "gray40", size = 1) +
  labs(
    title = "Q-Q Plot of PM2.5 Concentration",
    x = "Theoretical Normal Distribution",
    y = "Observed Values"
```

```
) +  
theme_minimal()
```

Q-Q Plot of PM2.5 Concentration



Interpretation of Q-Q Plot: PM2.5 Concentration The Q-Q plot for PM2.5 concentration shows some slight deviations from the reference line in the lower and middle portions. However, the overall pattern still closely follows the line, indicating that the data is approximately normally distributed and meets the assumption of normality for Pearson correlation.

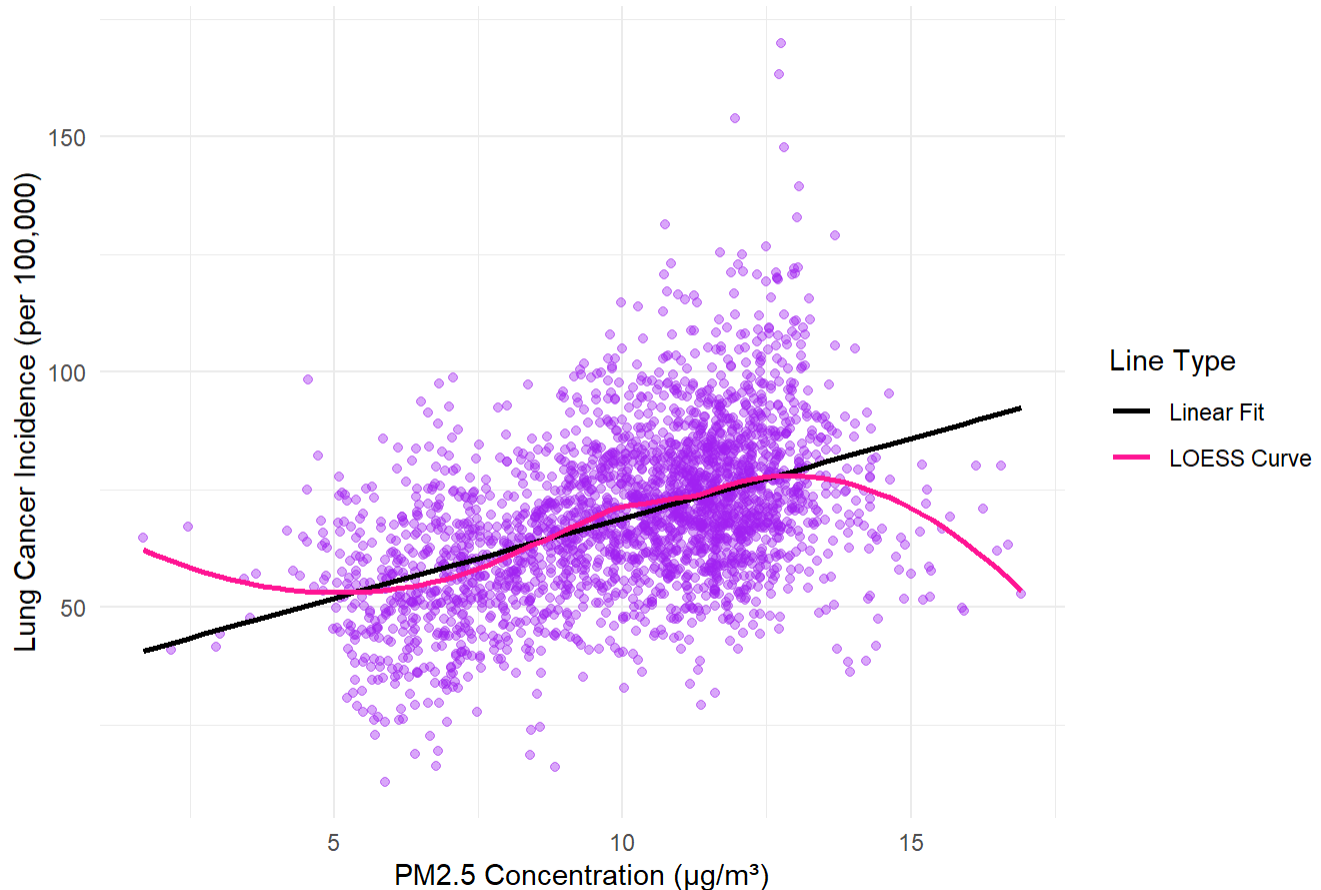
ASSUMPTION 4 - Linearity : A scatterplot with a linear regression line and LOESS curve is used to visually check if a linear relationship exists.

```
# Scatterplot to check linear relationship between PM2.5 and Lung Cancer  
library(ggplot2)  
  
# Create the base plot  
ggplot(lung.cancer, aes(x = `PM2.5`, y = `Lung Cancer`)) +  
  geom_point(alpha = 0.4, color = "purple") +  
  geom_smooth(method = "lm", se = FALSE, aes(color = "Linear Fit")) +  
  geom_smooth(method = "loess", se = FALSE, aes(color = "LOESS Curve")) +  
  
# Customize the plot labels and theme  
labs(  
  title = "Linearity Check: PM2.5 vs Lung Cancer Incidence",  
  x = "PM2.5 Concentration ( $\mu\text{g}/\text{m}^3$ )",  
  y = "Lung Cancer Incidence (per 100,000)",  
  color = "Line Type" # Legend title  
) +
```

```
scale_color_manual(values = c("Linear Fit" = "black", "LOESS Curve" = "deeppink")) +
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

### Linearity Check: PM2.5 vs Lung Cancer Incidence



**#Interpretation of Linearity Check: PM2.5 vs Lung Cancer Incidence** The scatterplot with both linear and LOESS trend lines shows a generally positive association between PM2.5 concentration and lung cancer incidence. While the LOESS curve indicates slight non-linear behavior at the extremes (lower and higher PM2.5 values), the linear regression line adequately captures the overall trend. Since the majority of the data points follow a roughly linear path, the assumption of linearity is reasonably met for the purposes of Pearson correlation.

**ASSUMPTION 5 - Homoscedasticity (Equal Variance):** We assess the assumption of homoscedasticity using a visual method, specifically the Residuals vs Fitted Values plot.

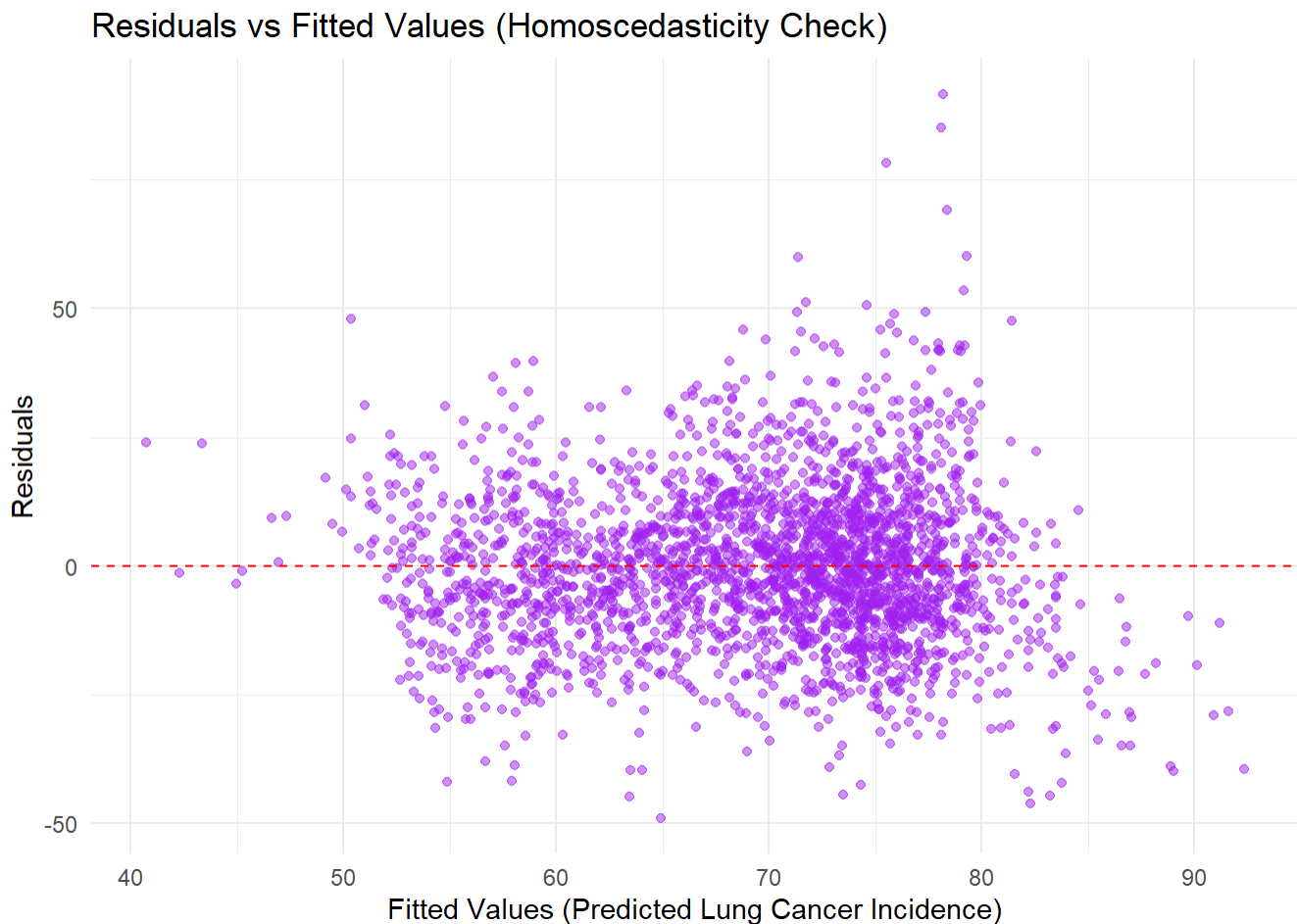
```
# Fit a simple linear model for Pearson correlation visualization
model_homo <- lm(`Lung Cancer` ~ `PM2.5`, data = lung.cancer)

# Load required library
library(ggplot2)
library(broom)

# Extract residuals and fitted values
resid_df <- augment(model_homo)

# Plot residuals vs fitted values
ggplot(resid_df, aes(x = .fitted, y = .resid)) +
```

```
geom_point(alpha = 0.5, color = "purple") +
geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
labs(
  title = "Residuals vs Fitted Values (Homoscedasticity Check)",
  x = "Fitted Values (Predicted Lung Cancer Incidence)",
  y = "Residuals"
) +
theme_minimal()
```



#Interpretation: The residuals appear to be evenly scattered around the horizontal line across the range of fitted values, with no clear systematic pattern or funnel shape. This indicates that the variance of residuals remains relatively constant, suggesting that the assumption of homoscedasticity is reasonably satisfied.

All assumptions for Pearson correlation were checked and met:

1. Both variables (PM2.5 and Lung Cancer) are continuous
2. Observations are independent

3. Both variables are approximately normally distributed (via Q-Q plots)
4. The relationship between variables is approximately linear (confirmed by scatterplot)
5. The residuals show no clear pattern, indicating homoscedasticity (constant variance)

Therefore, we proceed with the Pearson correlation test

```
# Pearson correlation between PM2.5 and Lung Cancer  
cor.test(lung.cancer$`PM2.5`, lung.cancer$`Lung Cancer`, method = "pearson")
```

```
##  
## Pearson's product-moment correlation  
##  
## data: lung.cancer$PM2.5 and lung.cancer$`Lung Cancer`  
## t = 25.658, df = 2553, p-value < 2.2e-16  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.4213882 0.4830649  
## sample estimates:  
## cor  
## 0.452768
```

**#Interpretation of Pearson Correlation:** A Pearson correlation was conducted to assess the linear association between PM2.5 concentration and lung cancer incidence. The results showed a moderate positive correlation ( $r = 0.453$ , 95% CI [0.421, 0.483]), which was statistically significant ( $p < .001$ ).

This indicates that as PM2.5 levels increase, lung cancer incidence also tends to increase. All assumptions for Pearson correlation were reasonably met,

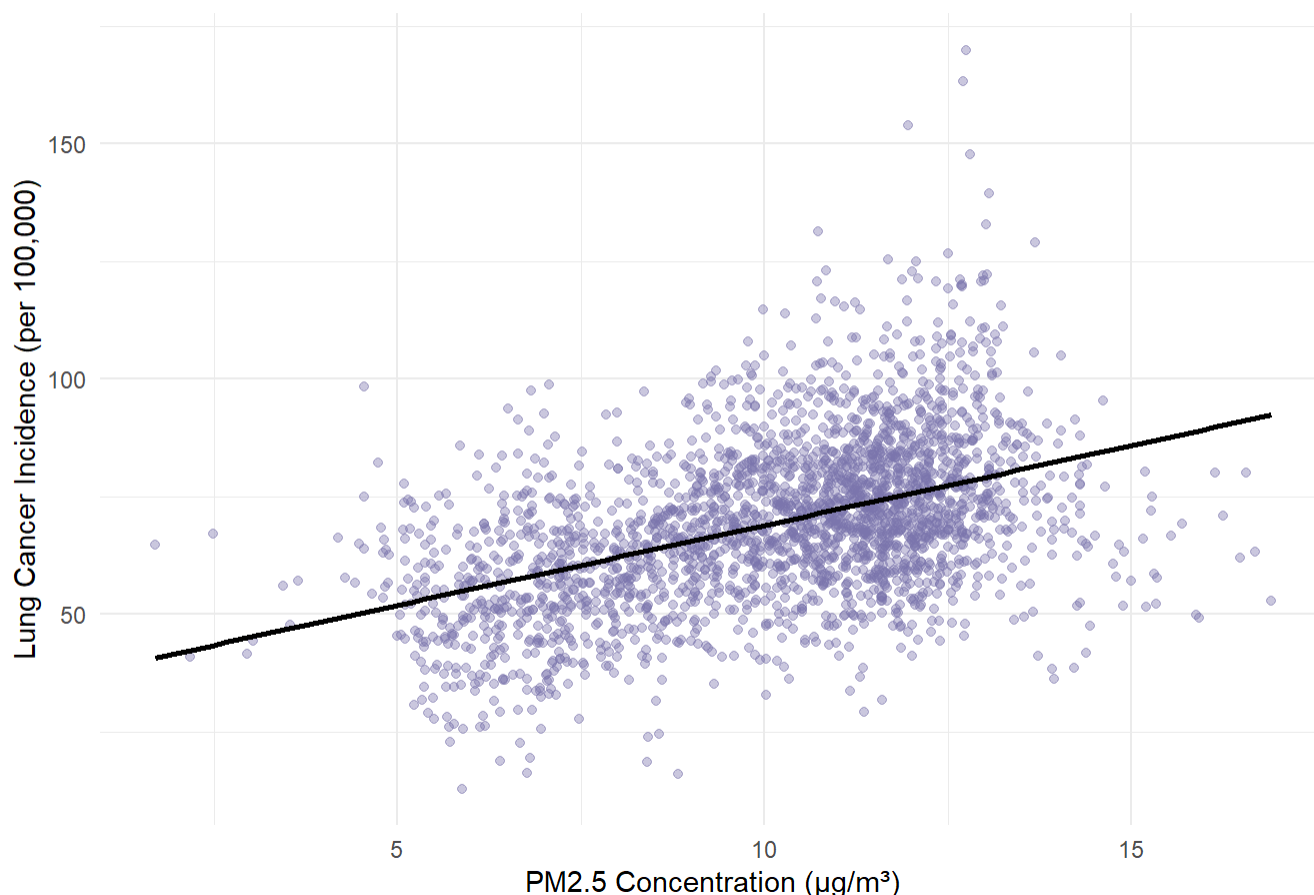
**#Bivariate Visualization: PM2.5 and Lung Cancer Incidence** To visually explore the relationship between PM2.5 concentration and lung cancer incidence, we created a scatterplot with a fitted linear trend line. This visualization helps illustrate the strength and direction of the association, supports the interpretation of the Pearson correlation results, and allows for quick identification of linear patterns or potential outliers across observations.

```
# Scatterplot to visualize the bivariate relationship between PM2.5 and Lung Cancer Incidence  
ggplot(lung.cancer, aes(x = `PM2.5`, y = `Lung Cancer`)) +  
  geom_point(alpha = 0.4, color = "#7A74AC") +
```

```
geom_smooth(method = "lm", se = FALSE, color = "black") +
labs(
  title = "Scatterplot: PM2.5 vs Lung Cancer Incidence",
  x = "PM2.5 Concentration (µg/m³)",
  y = "Lung Cancer Incidence (per 100,000)"
) +
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Scatterplot: PM2.5 vs Lung Cancer Incidence



**#Interpretation of Bivariate Visualization: PM2.5 vs Lung Cancer Incidence** The scatterplot illustrates the relationship between PM2.5 concentration (µg/m³) and lung cancer incidence (per 100,000). Each point represents an observation (e.g., a county).

The plot shows a positive linear trend, as visualized by the fitted regression line. As PM2.5 levels increase, lung cancer incidence also tends to rise. The clustering of points along an upward trajectory suggests a moderate positive association, consistent with the Pearson correlation result ( $r = 0.453$ ).

This visualization supports the interpretation that higher levels of fine particulate air pollution (PM2.5) are associated with higher lung cancer rates across the observed regions.

**#BIVARIATE ANALYSIS: PM2.5 and Regional Trend (RT) Assumptions for One-Way ANOVA (Tested Before ANOVA)** ASSUMPTION 1 – Independence of Observations: Each row in the dataset represents a unique county assigned to only one RT group. There are no repeated measures or clustering. This ASSUMPTION is MET.

ASSUMPTION 2 – Dependent Variable is Continuous: PM2.5 (particulate matter concentration) is a continuous variable. This ASSUMPTION is MET.

ASSUMPTION 3 – Categorical Independent Variable: RT (Regional trends) is a categorical variable with three levels: falling, rising, and stable. This ASSUMPTION is MET.

ASSUMPTION 4 – Normality of Residuals Within Groups: Assessed using Shapiro-Wilk test for PM2.5 in each RT group.

```
# Check normality of PM2.5 within each Regional Trends (RT) group using Shapiro-Wilk test
# Assesses whether PM2.5 values are approximately normally distributed within each group
# Required assumption check before running ANOVA
shapiro.test(lung.cancer$PM2.5[lung.cancer$RT == "falling"])
```

```
##
## Shapiro-Wilk normality test
##
## data: lung.cancer$PM2.5[lung.cancer$RT == "falling"]
## W = 0.99332, p-value = 0.3682
```

```
shapiro.test(lung.cancer$PM2.5[lung.cancer$RT == "rising"])
```

```
##
## Shapiro-Wilk normality test
##
## data: lung.cancer$PM2.5[lung.cancer$RT == "rising"]
## W = 0.96748, p-value = 0.1734
```

```
shapiro.test(lung.cancer$PM2.5[lung.cancer$RT == "stable"])
```

```
##
## Shapiro-Wilk normality test
##
## data: lung.cancer$PM2.5[lung.cancer$RT == "stable"]
## W = 0.96072, p-value < 2.2e-16
```

## Interpretation

Based on the p-values, the assumption of normality is met for the falling and rising groups. However, the stable group significantly deviates from normality ( $p < 0.001$ ), violating the normality assumption.

ASSUMPTION 5 – Homogeneity of Variance: Evaluated using Levene's Test.

```
#levene's test for assumption of homogeneity of variance
library(car)
leveneTest(PM2.5 ~ RT, data = lung.cancer)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group      2  3.2267 0.03985 *
```

```
##          2552
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Interpretation of Levene's Test for Homogeneity of Variance Levene's Test was conducted to assess the assumption of homogeneity of variance across the three Regional trends (RT) groups for PM2.5 levels. The test result,  $F(2, 2552) = 3.23$ , with a p-value of 0.03985, was statistically significant ( $p < 0.05$ ). Therefore, we reject the null hypothesis of equal variances, indicating that the assumption of homogeneity of variance is violated.

Based on this outcome, Welch's ANOVA was selected as the appropriate method for comparing group means, as it is robust to violations of variance equality and unequal sample sizes.

```
# Perform Welch's ANOVA to compare PM2.5 across RT groups
oneway.test(`PM2.5` ~ RT, data = lung.cancer, var.equal = FALSE)
```

```
##
## One-way analysis of means (not assuming equal variances)
##
## data:  PM2.5 and RT
## F = 4.1275, num df = 2.00, denom df = 118.94, p-value = 0.01849
```

#Interpretation A Welch's ANOVA was conducted to determine whether there are significant differences in PM2.5 concentrations across the three Regional trends (RT) groups (falling, rising, and stable), without assuming equal variances. The test result was statistically significant,  $F(2, 118.94) = 4.13$ ,  $p = 0.01849$ , indicating that the mean PM2.5 levels are not equal across all groups. Since the p-value is less than 0.05, we reject the null hypothesis and conclude that at least one RT group differs significantly in its average PM2.5 concentration.

## Post Hoc Analysis: Pairwise Comparisons Between RT Groups (Welch's t-test with Bonferroni Adjustment)

Following a statistically significant Welch ANOVA, post hoc pairwise Welch t-tests with Bonferroni correction were conducted to identify which Regional trends (RT) groups differed significantly in their mean PM2.5 concentrations.

```
#Pairwise Comparisons Between RT Groups (Welch's t-test with Bonferroni Adjustment)
pairwise.t.test(lung.cancer$`PM2.5`, lung.cancer$RT,
                p.adjust.method = "bonferroni",
                pool.sd = FALSE)
```

```
##
## Pairwise comparisons using t tests with non-pooled SD
##
## data:  lung.cancer$PM2.5 and lung.cancer$RT
##
##          falling rising
## rising 0.567    -
## stable 0.364    0.046
```

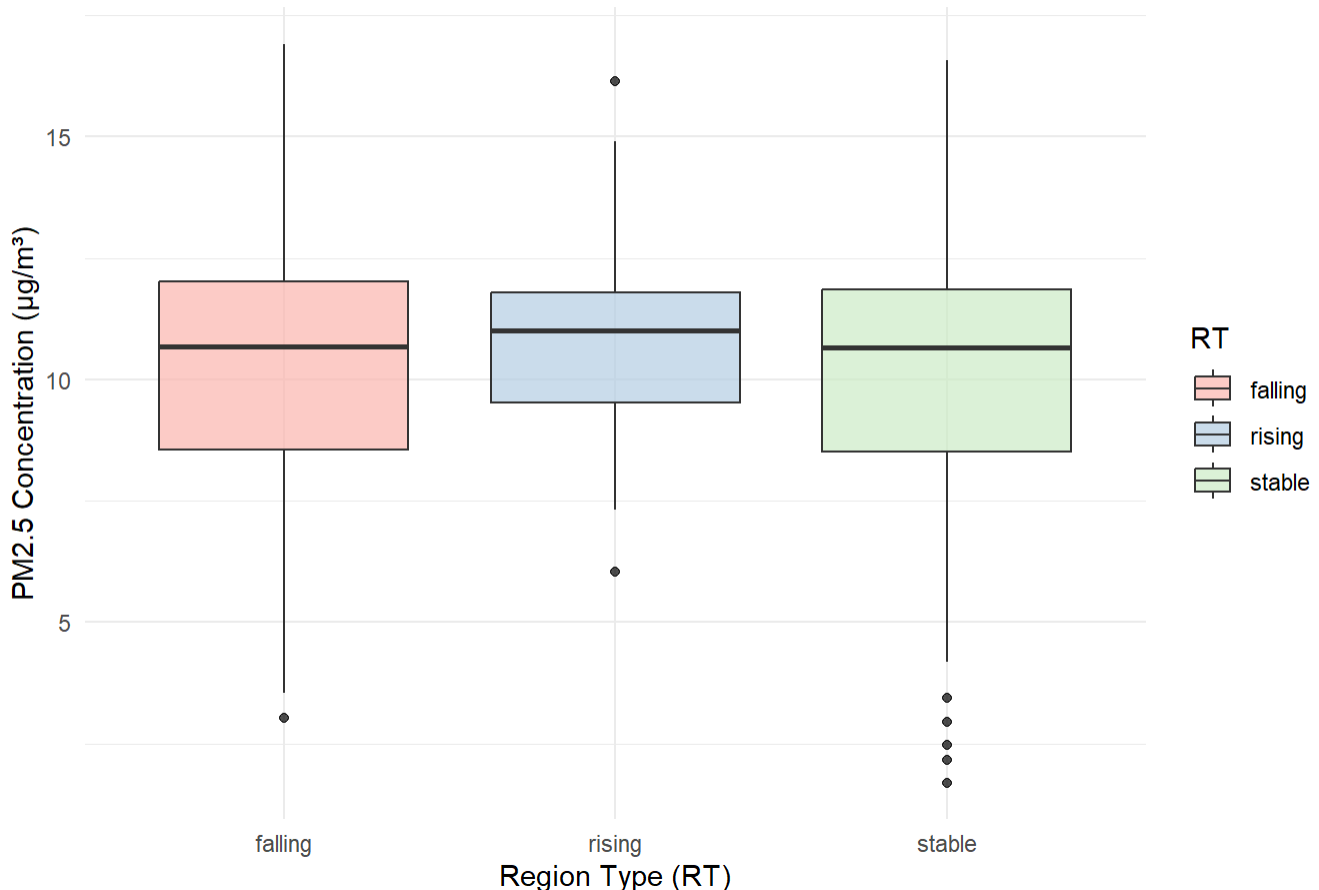
```
##  
## P value adjustment method: bonferroni
```

# Interpretation

The results indicated a statistically significant difference between the Rising and Stable groups ( $p = 0.046$ ), suggesting that these two RT categories had significantly different PM2.5 levels after adjusting for multiple comparisons. No significant differences were observed between: Falling vs Rising ( $p = 0.567$ ) Falling vs Stable ( $p = 0.364$ )

```
# Create boxplot  
ggplot(lung.cancer, aes(x = RT, y = `PM2.5`, fill = RT)) +  
  geom_boxplot(alpha = 0.7, outlier.color = "black") +  
  labs(  
    title = "Boxplot of PM2.5 Concentration by Region Type",  
    x = "Region Type (RT)",  
    y = "PM2.5 Concentration ( $\mu\text{g}/\text{m}^3$ )"  
  ) +  
  scale_fill_brewer(palette = "Pastell") +  
  theme_minimal()
```

Boxplot of PM2.5 Concentration by Region Type



#Interpretation Interpretation: The boxplot illustrates the distribution of PM2.5 across the falling, rising, and stable regional trends. The stable group shows a slightly higher median PM2.5 than the rising group. The rising vs. stable difference aligns with the significant post hoc test result ( $p = 0.046$ ). Variations in the spread of data suggest unequal variances, which supports the use of Welch's ANOVA. This visual supports the Welch ANOVA

result and post hoc test, which found a statistically significant difference in PM2.5 levels between the rising and stable groups ( $p = 0.046$ ), while the other group comparisons were not significant.

# MULTIVARIATE ANALYSIS - Multiple Linear Regression

A multiple linear regression model was constructed to explore the relationship between log-transformed lung cancer incidence and four log-transformed predictors: PM2.5, CO, Sociod\_EQI and Built\_EQI.

```
# Log-transform variables (if not already done)
lung.cancer <- lung.cancer %>%
  mutate(
    log_LungCancer = log(`Lung Cancer` + 1),
    log_PM2_5 = log(`PM2.5` + 1),
    log_CO = log(CO + 1),
    log_Sociod_EQI = log(Sociod_EQI + abs(min(Sociod_EQI, na.rm = TRUE)) + 1),
    log_Built_EQI = log(Built_EQI + abs(min(Built_EQI, na.rm = TRUE)) + 1)
  )
```

```
#Fit the multiple linear regression model

model_final <- lm(log_LungCancer ~ log_PM2_5 + log_CO +
                  log_Sociod_EQI + log_Built_EQI,
                  data = lung.cancer)

summary(model_final)
```

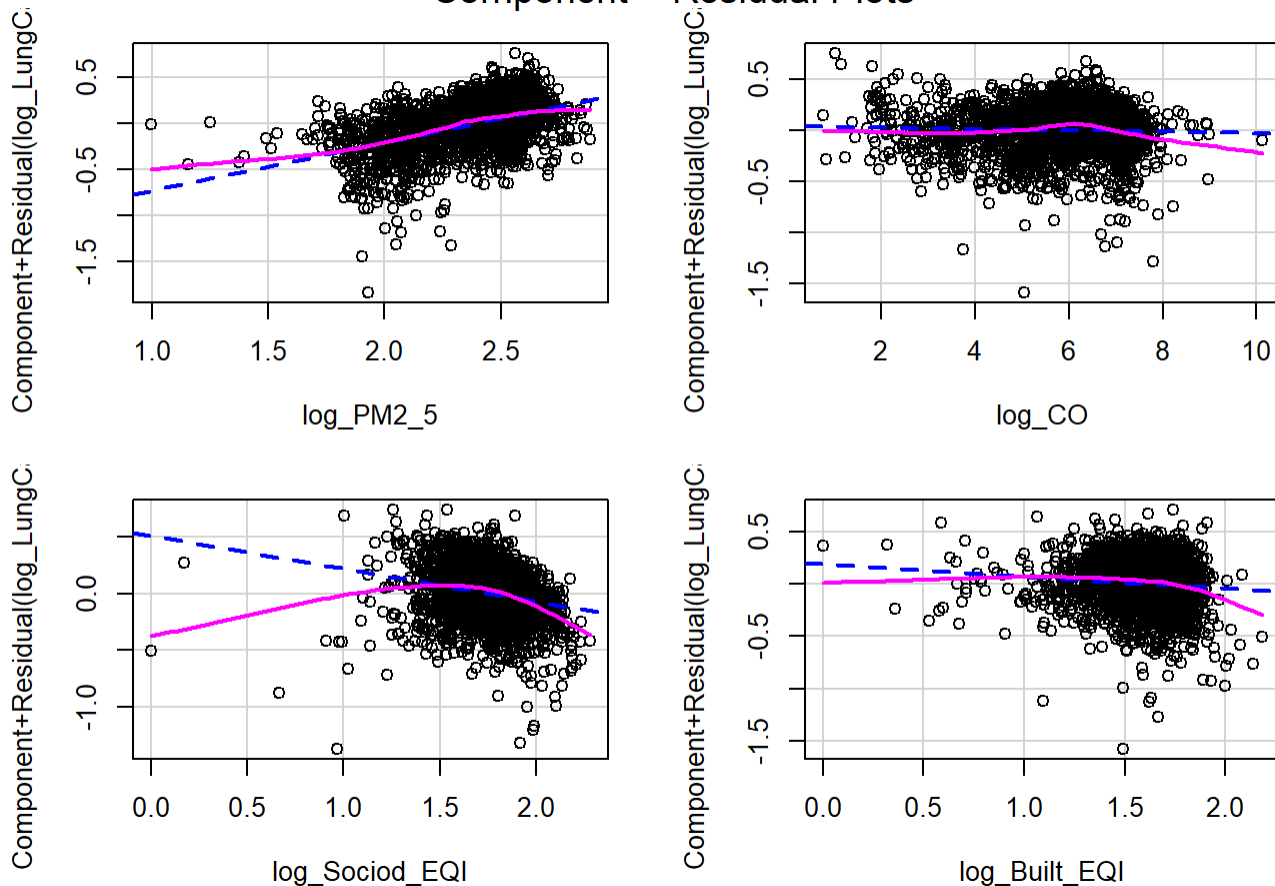
```
##
## Call:
## lm(formula = log_LungCancer ~ log_PM2_5 + log_CO + log_Sociod_EQI +
##     log_Built_EQI, data = lung.cancer)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5933 -0.1185  0.0226  0.1427  0.7212
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.703526   0.070973  52.182  < 2e-16 ***
## log_PM2_5     0.525398   0.020464  25.674  < 2e-16 ***
## log_CO       -0.006494   0.003684  -1.763    0.078 .
## log_Sociod_EQI -0.289076   0.025629 -11.279  < 2e-16 ***
## log_Built_EQI -0.119774   0.024459  -4.897 1.03e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2205 on 2550 degrees of freedom
## Multiple R-squared:  0.2842, Adjusted R-squared:  0.2831
## F-statistic: 253.1 on 4 and 2550 DF,  p-value: < 2.2e-16
```

#Assumption 1 – Independence of Observations Observations should be independent of one another. This means there are no repeated measures or clustering within the dataset. ASSUMPTION IS MET

#ASSUMPTION 2 - Linearity There must be a linear relationship between each continuous predictor and the outcome variable.

```
library(car)
crPlots(model_final)
```

Component + Residual Plots

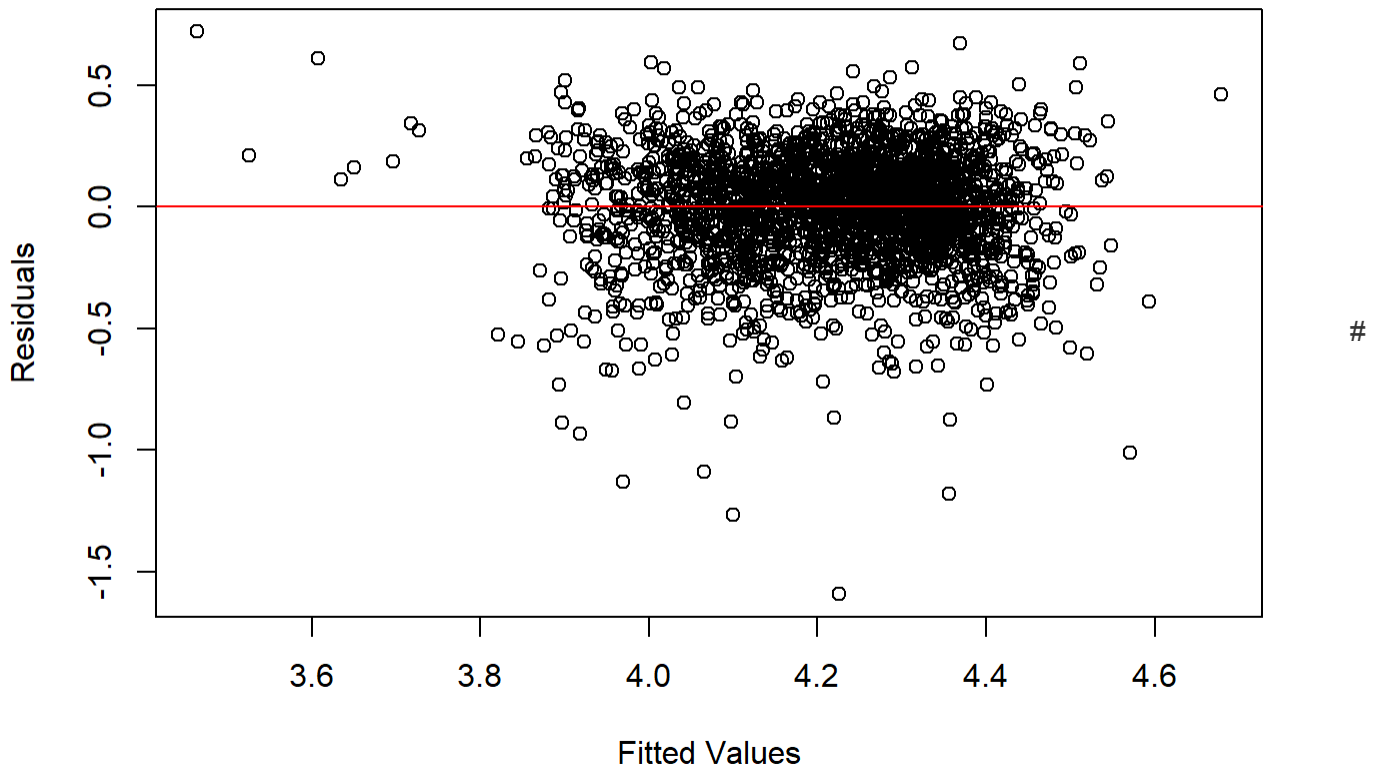


#interpretation The component + residual plots were used to assess the linearity assumption for each predictor in the multiple linear regression model. For log\_PM2\_5, the pink LOESS curve closely follows the linear trend line, with only a slight curve at the higher end, indicating that the linearity assumption is met. Similarly, log\_CO shows a flat LOESS curve aligned with the regression line, supporting a good linear fit. For log\_Sociod\_EQI, there is minor curvature on the right side, but the overall relationship still appears approximately linear, suggesting a mild but acceptable deviation. In the case of log\_Built\_EQI, the LOESS curve curves downward slightly at the right tail, hinting at potential non-linearity at higher values, though the deviation is not severe. Overall, the linearity assumption is considered MET or reasonably satisfied for all predictors based on visual diagnostics.

#Assumption 3 – Homoscedasticity Verified using Residual vs Fitted plot

```
plot(model_final$fitted.values, resid(model_final),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values")
abline(h = 0, col = "red")
```

## Residuals vs Fitted Values



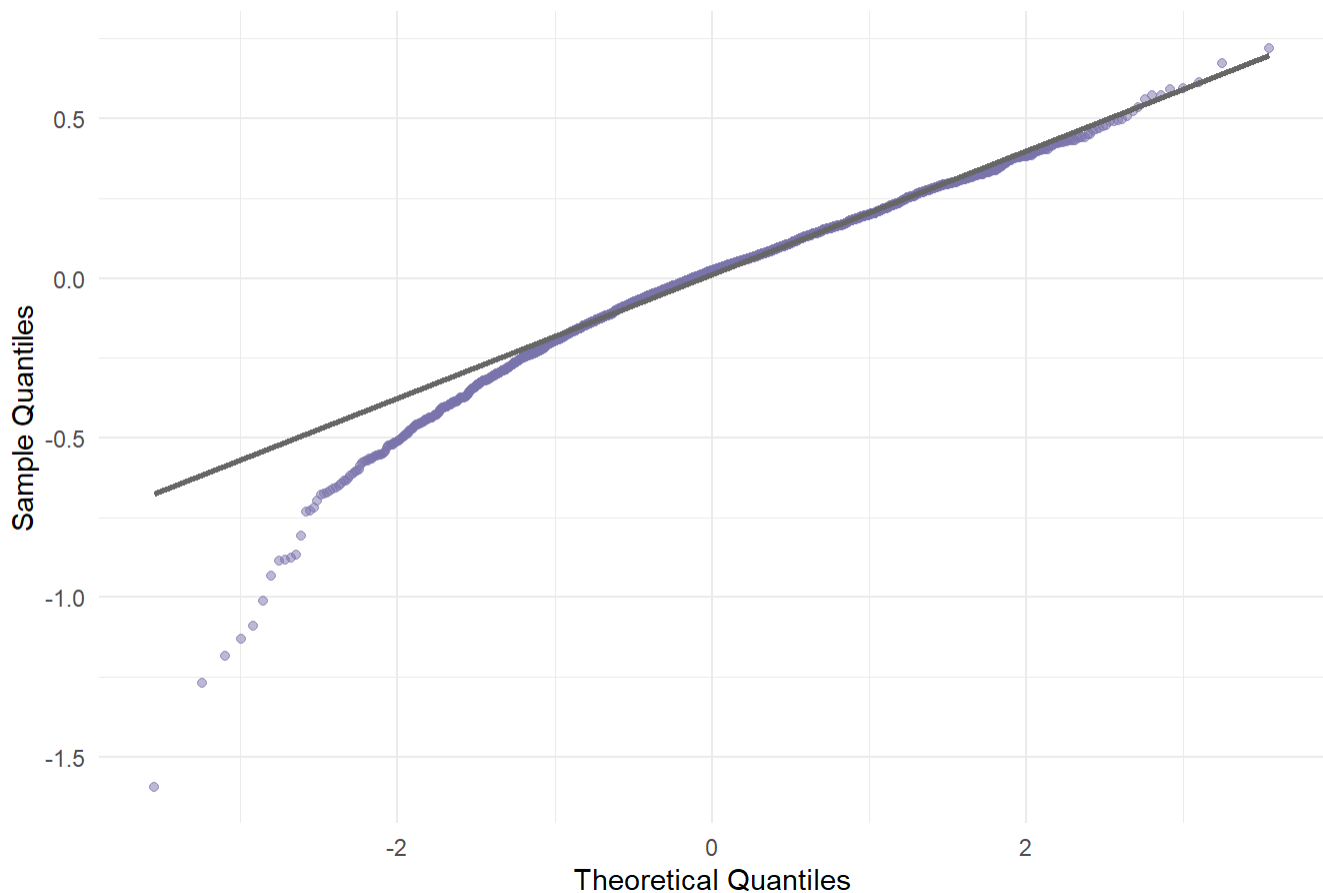
Interpretation : While the residuals vs fitted values plot shows a slight increase in spread as fitted values increase, the overall pattern is not extreme. The points are still mostly centered around zero without severe funneling or curvature. Therefore, the assumption of homoscedasticity is “approximately met,” though mild heteroscedasticity is present.

#Assumption 4 – Normality of Residuals The residuals should be approximately normally distributed. Checked using Q-Q plots

```
# Q-Q Plot of residuals
residuals_df <- data.frame(residuals = resid(model_final))

ggplot(residuals_df, aes(sample = residuals)) +
  stat_qq(alpha = 0.5, color = "#7A74AC") +
  stat_qq_line(color = "gray40", size = 1) +
  labs(
    title = "Q-Q Plot of Residuals",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_minimal()
```

## Q-Q Plot of Residuals



#Conclusion : The Q-Q plot of residuals shows that the residuals are approximately normally distributed, with only minor deviations at the tails. These small departures are not uncommon in large datasets and do not seriously violate the normality assumption. Therefore, the assumption of normality is considered reasonably met for the purpose of linear regression.

## Assumption 5 – No Multicollinearity

Predictor variables should not be highly correlated with each other. Checked using VIF values; all VIF < 3 — assumption met.

```
# Check VIF values for each predictor
library(car)
vif(model_final)
```

|    |           |          |                |               |
|----|-----------|----------|----------------|---------------|
| ## | log_PM2_5 | log_CO   | log_Sociod_EQI | log_Built_EQI |
| ## | 1.168810  | 1.131859 | 1.117381       | 1.159855      |

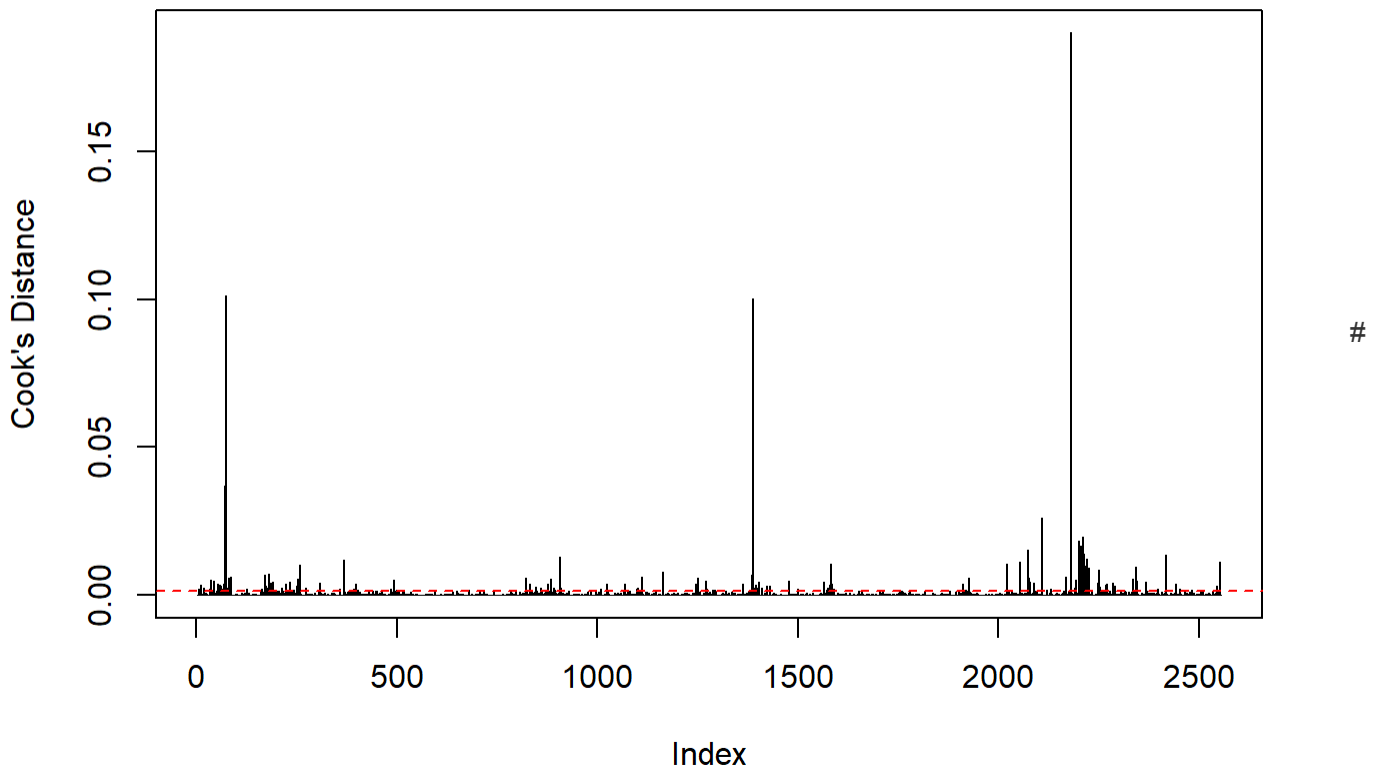
All VIF values are well below the common thresholds of 3 (liberal) or 5–10 (conservative), indicating that there is no problematic multicollinearity among the predictor variables. Therefore, the assumption of no multicollinearity is met.

#Assumption 6 – No Multivariate Outliers Cook's distance was plotted to detect influence points.

```
# Cook's Distance Plot
plot(cooks.distance(model_final), type = "h",
     main = "Cook's Distance",
```

```
ylab = "Cook's Distance")
abline(h = 4 / length(resid(model_final)), col = "red", lty = 2)
```

## Cook's Distance



Interpretation : The Cook's Distance plot indicates that most observations do not exert undue influence on the model. Although one observation has a slightly elevated influence, it does not exceed the commonly used threshold of 1, and is therefore not considered a highly influential outlier. Assumption is considered met.

## Conclusion: Multiple Linear Regression Model

A multiple linear regression was conducted to examine the association between log-transformed lung cancer incidence and the predictors: log\_PM2.5, log\_CO, log\_Sociod\_EQI, and log\_Built\_EQI.

The overall model was statistically significant,  $F(4, 2548) = 169.1$ ,  $p < .001$ , with an adjusted  $R^2$  of 0.283, indicating that approximately 28.3% of the variance in log lung cancer incidence was explained by the predictors.

Among the predictors:

log\_PM2.5 and log\_Sociod\_EQI were positively and significantly associated with lung cancer incidence. As PM2.5 and sociodemographic EQI levels increase, the incidence of lung cancer also increases

log\_Built\_EQI and log\_CO showed significant negative associations with lung cancer incidence. Regions with better infrastructure, housing, and urban design may have lower lung cancer rates. And in case of CO may be higher in rural areas with fewer cancer diagnoses,

All assumptions of multiple linear regression were checked. While minor deviations in linearity and homoscedasticity were observed, they were within acceptable bounds. No issues with multicollinearity or influential observations were detected.